

CWA SCIENCE CLASSES

Concept With Abhishek | Class 10 — Maths

Chapter 1: Real Numbers

1. Euclid's Division Lemma: For any two positive integers a and b , there exist unique integers q and r such that $a = bq + r$, where $0 \leq r < b$.
2. Euclid's Division Algorithm is used to find HCF of two positive integers.
3. Fundamental Theorem of Arithmetic: Every composite number can be expressed as a product of primes, and this factorisation is unique apart from the order of the prime factors.
4. $\text{HCF} \times \text{LCM} = \text{Product of the two numbers}$ (for two positive integers).
5. Proving irrationality: Use the fundamental theorem of arithmetic and proof by contradiction to show numbers like $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ are irrational.
6. Decimal Expansion: A rational number p/q has a terminating decimal expansion if $q = 2^m 5^n$; otherwise it is non-terminating repeating.